

Trend Estimation in the Forex and Commodity Market using the Fibonacci Sequence
Authors: Dr. Hanif Kanjer, Prof. Charmi Shah, Rahee Ambani
Rustomjee Business School

Corresponding Author: Rahee Ambani ;
Email ID : raheeambani@gmail.com ; Contact No. : +91 9819983636.

ABSTRACT

The paper has used the Fibonacci Sequence to obtain Fibonacci Retracement Ratios to study the changes in prices in the forex and commodities market. The first part of the paper lucidly explains the methodology behind the calculations of the fibonacci retracement ratios for a given set of data. Then, detailed study in price changes in USD vs. INR has been undertaken and fibonacci retracement ratios have been subsequently obtained. The study indicates that the predictions using retracement ratios are more precise for shorter spans of time. The paper then focuses finding the numerical value of the precision provided by the fibonacci retracement ratios. It has been calculated by taking into account data of sugar commodity prices over the last five years in India.

INTRODUCTION

The application of Fibonacci Retracement Ratios is a relatively new idea and the paper has attempted to further the understanding and application of these ratios in technical analysis of forex and commodities. Osler (2000) and Osler (2003) posit that the predictive power of fibonacci elements has a self-fulfilling component.

This paper has attempts to apply the Fibonacci Retracement Ratios to forex, specifically USD vs. INR. USA is often looked upon as one of the most powerful nations in the world, and India is one of the fastest developing democracies. Any economic, social and political changes in either one of the nation's tends to have ramifications on the rest of the world. Hence, the USD vs. INR is very important and adequately representative of the rest of the forex.

The paper has also looked into applying Fibonacci Retracement Ratios to the price of sugar commodity in India. India, mainly an agro-based nation, produces over 336 million tonnes of sugarcane and 25 million tons of sugar each year. The production of sugar over the last ten years has been increasing at CAGR of 2.63%. Hence, sugar commodity plays a vital role in the economy of the nation and is a sufficient indicator of the growth and value of other similar commodities.

LITERATURE REVIEW

Murphy (1999) believes that anything that can possibly affect the price ... is actually reflected in the price of that market. J.M. Hurst (1973), in his "cycle courses" referred to Harmonic Trading. His principle of harmonicity that states: "The periods of neighboring waves in price action tend to be related by a small whole number." Fibonacci numbers and price patterns may manifest these relationships and provide an adequate method to determine where the turning points will occur in significant trends. Adam Grimes(2011), author of "The Art and Science of Technical Analysis" found that there was a closest fit to 65% with a standard deviation of 20.7%.

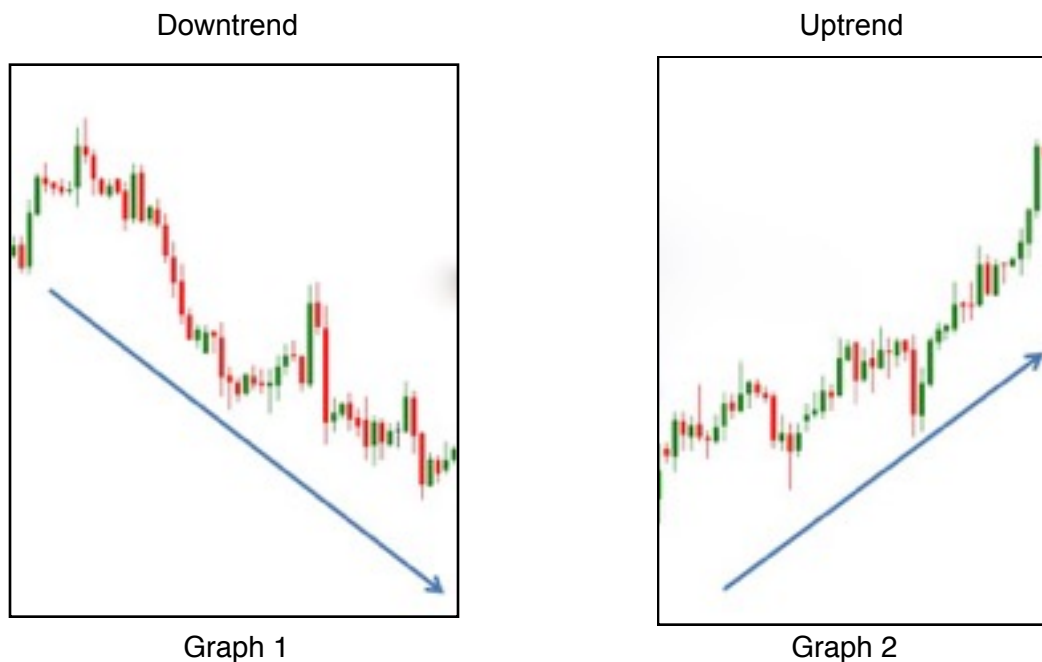
An Italian mathematician Leonardo Fibonacci, in his book Liber Abaci, introduced the idea of a sequence of numbers where each number in the sequence is the sum of its previous two values. Today, it is famously known as the Fibonacci Sequence and scientists have studied it's wide range of applications : in higher level mathematics, physics and biology.

For any nth term F_n , the Fibonacci Sequence is :

F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_n
0	1	1	2	3	5	8	13	21	...

Trading is a game of probabilities. It is every traders aim be to find opportunities which have higher probabilities of success. Hence, recently, many technical analysts have started looking at using the Fibonacci Sequence during their analysis. The ratios of consecutive numbers in the sequence is found, and these ratios are used to predict the values of stocks, commodities and forex.

The Fibonacci Retracement Ratios give traders levels or values at which a possible trend reversal can occur. The general assumption using Fibonacci Retracements is: an initiated trend will pursue its direction. The time frame used for the same is dependant on the traders preferences; while daily traders may find suitable ratios using hourly data, long term investors find the fibonacci retracement ratios using data of over five years. In order to find suitable ratios, one has to first identify the current trend of the stock/forex/commodity. The toughest task is to identify a working swing high or swing low. There are majorly two kinds of trends : uptrends and downtrends. An uptrend in technical analysis can be defined as a series of higher highs and higher lows. A downtrend can be defined as a series of lower highs and lower lows. Below the graphic representation of both in a candlestick chart has been given respectively.



Algorithmically, a Fibonacci number f_n is given by the following recurrence:

$$f_0 = 0, f_1 = 1, \\ f_{n+1} = f_{n-1} + f_n, n \geq 1.$$

Every number in this sequence is approximately 1.618 times greater than the preceding number. This common relationship between every number in the series is the foundation of the common ratios used in retracement studies.

Fibonacci ratios can be defined mathematical relationships, expressed as ratios, derived from the Fibonacci sequence. These numbers are used to create levels which can be used to study and predict successive highs or lows.

These levels are created by drawing a trendline between two extreme points and then dividing the vertical distance by the key Fibonacci ratios of 23.6%, 38.2%, 50%, 61.8% and 100%.

The key Fibonacci ratios are 0%, 23.6%, 38.2%, 61.8%, and 100%

Fibonacci Retracement Level	Formula	Approximate Value
F _{100%}	$(0.5 + 0.5\sqrt{5})^0$	1
F _{61.8%}	$(0.5 + 0.5\sqrt{5})^{-1}$	0.618034
F _{38.2%}	$(0.5 + 0.5\sqrt{5})^{-3}$	0.236068
F _{0%}	$(0.5 + 0.5\sqrt{5})^{-\infty}$	0

Table 1

Similarly, other Fibonacci ratios including F_{76.4%}, F_{50%}, F_{138.20%}, F_{161.8%} can be obtained and used to analyse the data more efficiently.

We know that Fibonacci retracement levels can be used to forecast potential areas of support or resistance at which traders can enter the market with a view to catching the resumption of an initial trend. Fibonacci extensions can compliment this strategy by giving traders Fibonacci based profit targets.

Fibonacci extensions consist of levels drawn beyond the standard 100% level and can be used by traders to project areas that make good potential exits for their trades in the direction of the trend. The major Fibonacci extension levels are 161.8%, 261.8% and 423.6%.

To use the fibonacci retracement ratios in technical analysis, the following methodology has been used.

1) Fibonacci Retracement Ratios are calculated seperatly for an uptrend and a downtrend. For each, retracement and extension values are calculated.

Fibonacci retracement levels are those that are lower than the 100% of a price swing, while extensions are those that are above 100%. Fibonacci levels are used as support or resistance, and/or for projection of profit targets. In an uptrend, the traders and investors should attempt to earn profit on a long trade at these extension levels where as in a downtrend the traders and investors should attempt to earn profit on the short trade.

2) To calculate the Fibonacci retracement ratio values for an Uptrend.

Let H be the highest value over a period of time and L be the lowest value for the same period.

The formula for retracement values is given as:

Retracement = $H - (H - L) (0, 0.236, 0.382, 0.5..)$

The formula for extension values is given as:

Extension = $H + (H - L) (2.618, 2, 1.618, 1.382...)$

3) To calculate Fibonacci retracement ratio values for a Downtrend

Let H be the highest value over a period of time and L be the lowest value for the same period.

The formula for retracement values is given as:

Retracement = $L - (H - L) (0, 0.236, 0.382, 0.5..)$

The formulae for extension values is given as: \

Extension = $L + (H - L)(2.618, 2, 1.618, 1.382...)$

The above values are represented graphically along with prices of data used. The retracement and extension values are horizontal lines parallel to the x-axis, plotted according to their corresponding values on the y-axis. The data points are connected by a line, i.e. a line graph is plotted.

Fibonacci retracements use horizontal lines to indicate areas of support or resistance at the key Fibonacci levels before it continues in the original direction. These levels are created by drawing a trend line between two extreme points and then dividing the vertical distance by the key Fibonacci ratios of 23.6%, 38.2%, 50%, 61.8% and 100%.

RESEARCH METHODOLOGY

The research was quantitative in nature and data was collected from reliable secondary sources. The following data was collected from January, 2011 to April, 2016 on a monthly and weekly basis respectively

1. American Dollar vs Indian Rupee
2. Sugar trading commodity prices

Data analysis was done using Descriptive Statistics (Mean, median, mode, standard deviation, percentages, range etc.) The data was graphically represented and the observations were made on the same.

RESEARCH OBJECTIVE

- 1) To observe the efficiency of Fibonacci Retracement ratios in forex across different time spans
- 2) To observe the efficiency of Fibonacci Retracement ratios in commodities over a span of 5 years

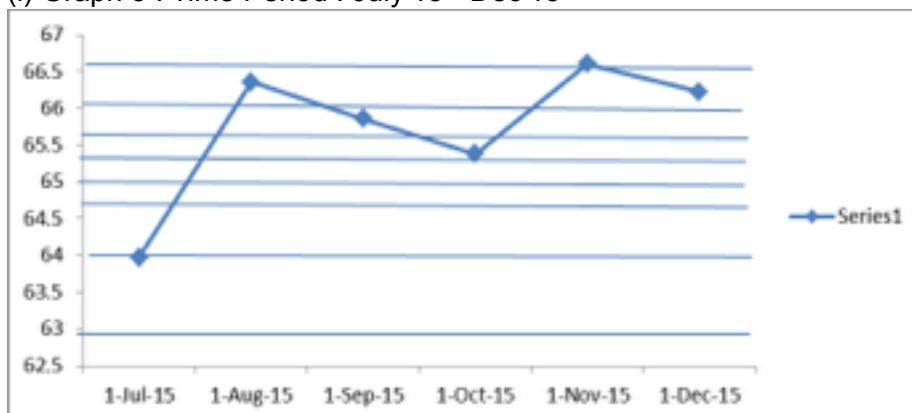
OBSERVATIONS

- 1) To observe the efficiency of fibonacci retracement ratios in forex across different time spans. To obtain the objective, USD vs INR has been taken into consideration over a period of two and a half years.

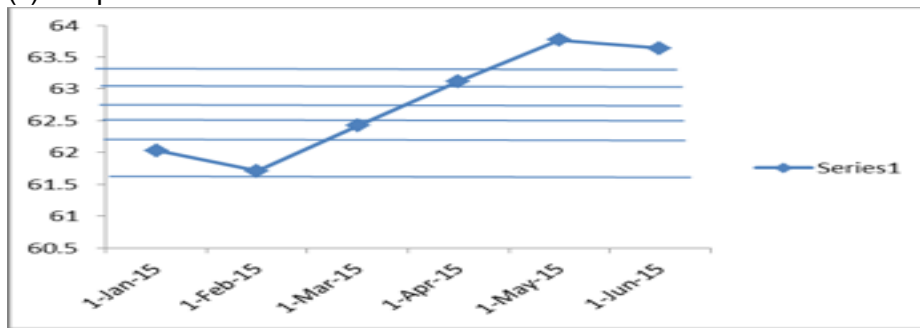
Methodology: The period of July, 2013 to December, 2015 has been divided into 5 equal segments, each six months long. Each segment has been considered separately for observation. Within each segment, the monthly and weekly prices of USD xs INR are obtained. The retracement ratio values are obtained for the monthly data using the previously derived formulae. Then the data and the ratio lines are plotted to obtain a precise graphic representation of the data. Similarly, the weekly prices of USD vs INR are obtained for a period of six months. Subsequently, the retracement ratios are calculated and then plotted on a graph. Below, the aforementioned graphs have been input.

A) Monthly data

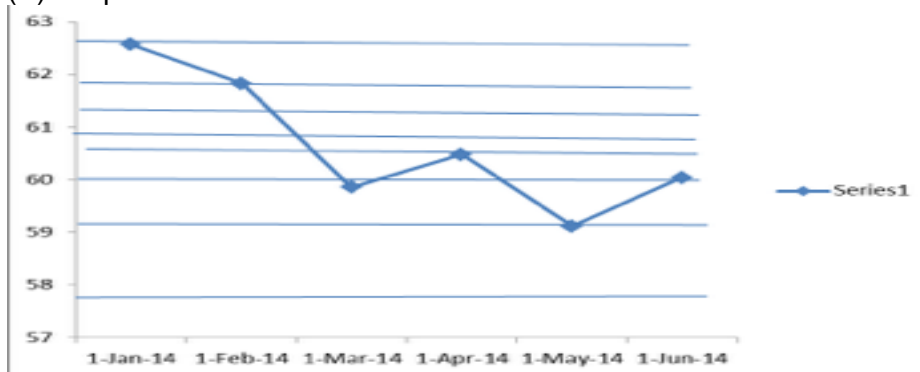
(i) Graph 3 : Time Period : July'15 - Dec'15



(ii) Graph 4 : Time Period : Jan'15 - June'15

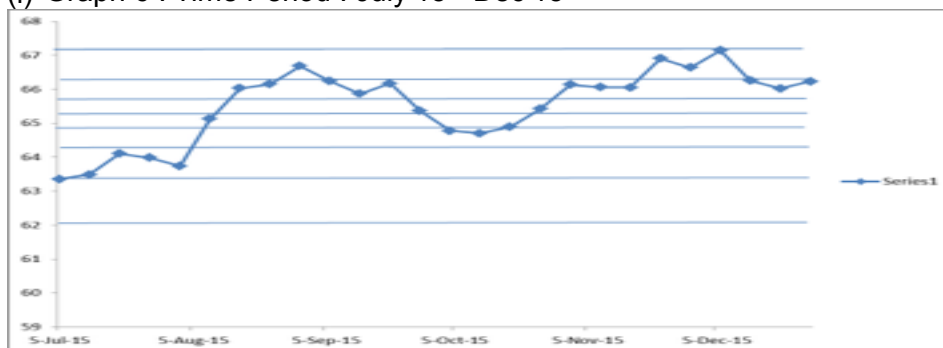


(iii) Graph 5 : Time Period : Jan'14 - June'14

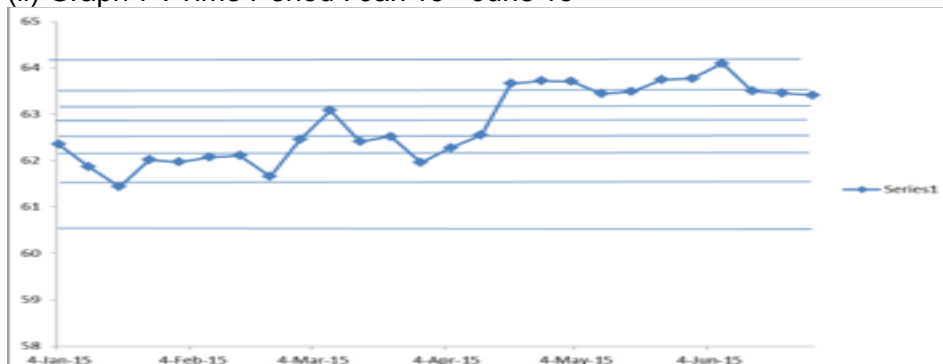


B) Weekly data

(i) Graph 6 : Time Period : July'15 - Dec'15



(ii) Graph 7 : Time Period : Jan'15 - June'15



(iii) Graph 8 : Time Period : Jan'14 - June'14



Observations: Studying the above graph in detail, the conclusion has been reached that in general, the highs and lows of the weekly data are closer in value to the retracement ratio levels as that compared to those of the monthly data.

Essentially, it is more profitable to predict values for weekly data using fibonacci retracement ratios than to use the same for monthly data.

The loss of precision in monthly data could be due to external socio-economic forces. A week is a very short time for socio-economic factors to be able to have a large impact on prices.

However, a month is enough time for the factors to be able to influence the prices. Factors such as trade deals between nations, new laws, policy changes in either of the two countries, politics and government aims and policies, crude prices, etc. have a large impact on the exchange rate of countries.

2) To observe the efficiency of fibonacci retracement ratios in commodities over a span of 5 years, sugar has been taken into consideration. The time period is May, 2011 to May, 2016. The precision for the same has been obtained.

Methodology: The time period has been divided into 5 segments, each 1 year long. Each segment has been considered independently. For each segment, the monthly prices of sugar have been obtained. Using the data, the fibonacci retracement levels are calculated. This is followed by the graphic representation of the data points and the retracement levels.

To obtain the precision: Detailed observation of the graph is undertaken. The highs and lows in the graph are taken into consideration. Points where a high or low exactly coincides with the retracement levels is noted down. Hence, the precision levels are obtained.

Note: Below, a few of the segments that have been studied are shown. The bold line trend indicates the sugar prices over that time period. The horizontal lines are the fibonacci retracements. The Retracement and Extensions table to the right of each graph shows the individual fibonacci retracement values at each level.

A) Graph 8 : Time Period : May'11 to April'12



Table 2

Retracements		Extentions	
138.20%	31.6208	61.80%	17.1585
100% (a)	28.8585	100%	14.3962
76.40%	27.152	138.20%	11.6339
61.80%	26.0962	161.80%	9.9273
50%	25.243	200%	7.165
38.20%	24.3897	261.80%	2.6962
23.60%	23.3339		
0% (b)	21.6274		

B) Graph 9 : Time Period : May'14 to April'15



Table 3

Retracements		Extentions	
138.20%	27.5222	61.80%	13.2804
100% (a)	24.802	100%	10.5602
76.40%	23.1215	138.20%	7.8399
61.80%	22.0818	161.80%	6.1594
50%	21.2416	200%	3.4392
38.20%	20.4013	261.80%	-0.9615
23.60%	19.3616		
0% (b)	17.6811		

C) Graph 10 : Time Period : May'15 to April'16



Table 4

Retracements		Extentions	
0% (b)	22.8179	261.80%	42.4994
23.60%	21.0437	200%	37.8534
38.20%	19.9461	161.80%	34.9816
50%	19.059	138.20%	33.2074
61.80%	18.1719	100%	30.3356
76.40%	17.0743	61.80%	27.4638
100% (a)	15.3001		
138.20%	12.4283		

Note : The precision data shown in table 1 below show the probability of a high or low to be precisely on the fibonacci retracement levels. However, it must be noted, that looking at the graphs above, the highs and lows are generally just above or below a retracement level. Hence, a possible precision of 0.6 does not indicate that the probability of a low being far away from a retracement level is 0.4. Instead, it means that 60% of the times, the high or low will be exactly on the retracement levels and 40% of the time the high or low will be around retracement level value.

Table 5

Date	Precision
May'11 - May'12	0.57
May'12 - May'13	0.67
May'13 - May'14	0.5
May'14 - May'15	0.5
May'15 - May'16	0.6

Observations : The average precision of fibonacci retracement ratios over a period of 5 years for sugar commodity in India is 0.57. This means that the probability of a high or low for sugar prices in India to lie on a retracement level is 0.57. Looking at the graphs(11-15) we can infer that the highs and lows lie close to the retracement levels. Hence, this method can be useful for corporations or entities looking to make a profit by trading commodities in India.

LIMITATIONS OF STUDY

1. While carrying out the study the only forex used was USD vs INR and only commodity used was sugar. The results obtained for the above may not be the results obtained over a wide range of other forex or commodities.
2. The time span used was two and a half years and five years for the first and second observation respectively. This is a relatively short time span, and the values, trends and observations could differ over a longer period of time.

Fibonacci Retracement Ratios is a relatively esoteric field, and has much more scope in the future.

1. Can be applied to different commodities and forex
2. It can help investors leverage on arbitrage opportunities
3. Further research can be carried out to make the application simpler and more precise
4. Research on applications of these ratios in other fields such as medicine can be looked into

REFERENCES

- Mark Whistler, May 12, 2010, How to Profit from Fibonacci Retracements in Forex Trading, [online], [read in 10 february, 2011],
- MFTA Pershikov, Viktor (2014). *The Complete Guide To Comprehensive Fibonacci Analysis on FOREX*. ISBN 978-1607967606.
- Stevens, Leigh (2002). *Essential technical analysis: tools and techniques to spot market trends*. New York: Wiley. ISBN 0-471-15279-X. OCLC 48532501.
- Brown, Constance M. (2008). *Fibonacci analysis*. New York: Bloomberg Press. ISBN 1-57660-261-3.